

**COURSE PLAN**

Department :	MATHEMATICS				
Course Name & code :	Elementary Number Theory			MAT 2137	
Semester & branch :	III SEMESTER		Mathematics and Computing		
Name of the faculty :	Dr Vadiraja Bhatta G. R.				
No of contact hours/week:	L	T	P	C	
	2	1	0	3	

COURSE OUTCOMES (COS)

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	PSO	BL (Recommended)
CO1	Construct models for solving problems using basic concepts of number theory.	10	8			1, 2
CO2	Apply the fundamentals of number theory to solve advanced number theoretic problems.	6	10			1, 3, 4
CO3	Evaluate number theoretic functions applicable for factoring large integers.	8	12			2,3,4,5
CO4	Analyse cryptographically significant concepts using number theory.	5	10			4, 5
CO5	Apply the number theoretic concepts, methods and functions to cryptography.	7	10			3, 4
	Total	36	50			

***** COURSE LEARNING OUTCOMES (CLOS)**

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes(POs)	Learning Outcomes (LOs)	BL (Recommended)
CLO1	Construct models for solving problems using basic concepts of number theory.	10	8			1, 2
CLO2	Apply the fundamentals of number theory to solve advanced number theoretic problems.	6	10			1, 3, 4
CLO3	Evaluate number theoretic functions applicable for factoring large integers.	8	12			2,3,4,5
CLO4	Analyse cryptographically significant concepts using number theory.	5	10			4, 5
CLO5	Apply the number theoretic concepts, methods and functions to cryptography.	7	10			3, 4
Total		36	50			

*** Applicable to programs applied for IET accreditation only.

Assessment Plan

IN – SEMESTER ASSESSMENTS

S. No.	Assessment Mode		Assessment Method	Time Duration	Marks	Weightage	Typology of Questions (Recommended)	Schedule	**Topics Covered
1	MISAC	1	Assignment 1	15 days	5	10 MCQs $\times \frac{1}{2} = 5$	Bloom's taxonomy (BT) level of the question should be L3 and above.		L1—L10
		2	Assignment 2	15 days	5	2 STQ $\times 2\frac{1}{2}= 5$	Bloom's taxonomy (BT) level of the question should be L3 and above.		L11—L20
		3	Mid-semester Exam	60 Mins	30	Objective: 5M 10 MCQs $\times \frac{1}{2} = 5$ marks Descriptive: 25 Marks (3 Questions of 2 marks + 5 Questions of 3 marks+ 1 question of 4 Marks)	Bloom's taxonomy (B) level of the question should be L3 and above.		L1 – L15
		4	Assignment 3	15 days	5	10 MCQs $\times \frac{1}{2} = 5$	Bloom's taxonomy (BT) level of the question should be L3 and above.		L21—L26
		5	Assignment 4	15 days	5	2 STQ $\times 2\frac{1}{2}= 5$	Bloom's taxonomy (BT) level of the question should be L3 and above.		L27—L34
<u>END – SEMESTER ASSESSMENT</u>									

1	Regular/Make-Up Exam	180 Mins	50	Answer all 5 full questions of 10 marks each. Each question can have 3 parts of 2/3/4/5/6 marks.	Bloom's taxonomy (BT) level of the question should be L3 and above.	17 th week of the semester	Comprehensive examination covering full syllabus.
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***** Individual faculty will be entering the topics***

****** Individual faculty must identify the assessment method from table 3 and fill in the details.***

NOTE: Information provided in the table is as per the In-semester assessment plan and schedule of V and VII semester B. Tech provided from Academic Section.

 after midsems

 cryptography

 till midsems

LESSON PLAN

L No	TOPICS	Course Outcome Addressed
1	Divisibility- properties of divisibility	1
2	Division Algorithm – related problems	1
3	The Greatest Common Divisor	1
4	Euclidean Algorithm- Finding GCD	1
5	Gaussian Integers – GCD of Gaussian Integers.	1
6	Primes as a Sum of Two squares.	1
7	The Theory of Congruences - Examples	1
8	Basic properties of Congruences	1
9	Linear Congruences and Chinese Remainder Theorem	1
10	Problems on Chinese Remainder Theorem	1
11	The Diophantine Equation	2
12	Solution of Diophantine equations.	2
13	Prime numbers and their distributions	2
14	Fundamental Theorem of Arithmetic	2
15	Some more problems on prime numbers	2
16	Wilson's Theorem and Problems	2
17	Fermat's Little Theorem and problems	3
18	Euler's Phi-function and properties	3
19	Properties of Euler's phi-function	3
20	Euler's Theorem and problems on Euler's theorem	3
21	Some more problems on Euler's phi- function	3
22	Number Theoretic functions	3
23	Greatest Integer Function and Application	3
24	Problems on Greatest Integer Function	3
25	Quadratic Reciprocity – Introduction	4
26	Euler's Criterion	4
27	Legendre symbol and properties of Legendre symbol	4
28	Evaluation of Legendre Symbols	4
29	Problems on law of quadratic reciprocity	4
30	Square root mod p	4
31	Factoring Large Integers.	4
32	Cryptography- Introduction.	5
33	Basic cryptosystems- problems	5
34	Problems on Basic cryptosystems	5
35	Discrete log problem	5
36	The Knapsack problem and Knapsack Cryptosystem	5

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3						1				1			
CO2	3	3						1				1			
CO3	2	3						1				1			
CO4	2	2						1				1			
CO5	2	1						1				1			
Articulation Level	2.4	2.4						1				1			

FACULTY MEMBER TEACHING THE COURSE: Dr Vadiraja Bhatta G. R.

References:

1. David M. Burton- Elementary Number Theory, 6th Edition, Tata Mc Graw Hill, 2007
2. Neal Koblitz - A Course in Number Theory and Cryptography, Springer, second Edition.
3. Neville Robbins- Beginning Number Theory, Narosa Publishing House, 2007.
4. Tom M. Appostal- Introduction to Analytic Number Theory, Springer, 1976

Submitted by: Dr. Vadiraja Bhatta G R

(Signature of the faculty)

Date: 21.7.2025

Approved by: Dr Kuncham Syam Prasad.

(Signature of HOD)